Jiace Wu

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EDUCATION

Hangzhou Dianzi University, Hangzhou, China *Sep 2021 - Jun 2024*

Master of Engineering in Mechanical Engineering

- **GPA**: 86.5/100
- *Relevant Coursework*: Numerical Analysis, Ultrasonic Machining Theory and Technology, Theory and Technology of Mechanical Manufacturing Methods, Machining Theory and Mechanical Manufacturing Equipment

Jingdezhen Ceramic University, Jingdezhen, China *Sep 2017- Jun 2021*

Bachelor of Engineering in Mechanical Design, Manufacturing and Automation

- **GPA**: 81.5/100
- *Relevant Coursework*: Materials Mechanics, Engineering Materials, Mechanical Precision Design, Finite Element Method, Manufacturing Methods

PUBLICATIONS

- **Wu, J.**, Hu, X., Ji, H., Zhu, Q., & Hong, W. (2025). Study on cutting force of ultrasonic cutting disc cutter for Nomex honeycomb material. *The International Journal of Advanced Manufacturing Technology*, *139*, 4339–4351. (Q2 Journal). https://doi.org/10.1007/s00170-025-16166-7
- **Wu, J.**, Hu, X., & Ji, H. (2024). Study of equivalent mechanical parameters out-of-plane of wave-absorbing honeycomb panel and core layer. *Journal of Hangzhou Dianzi University: Natural Sciences, 44*(3), 94–102. (Chinese Core Journal). https://doi.org/10.13954/j.cnki.hdu.2024.03.013

RESEARCH INTERESTS

Composite Materials, Ultrasonic Machining, Cutting Mechanism, Damage and Fracture Mechanisms of Materials

RESEARCH EXPERIENCE

Study on Ultrasonic Machining Stability of Wave-Absorbing Honeycomb Materials *Jun 2019 - Jun 2020*

National Natural Science Foundation of China (NSFC) Project

- Performed finite element simulation of ultrasonic cutting for wave-absorbing honeycomb composites using Abaqus.
- Developed machine vision-based contour recognition and binarization processing in MATLAB to extract and fit dimensional profiles of materials.
- Investigated fracture behavior of composites by analyzing micro-crack propagation under conventional vs. ultrasonic vibration-assisted cutting.
- Established a cutting force model incorporating ultrasonic amplitude, frequency, feed rate, and disc speed, integrating parameters into a mathematical framework to elucidate factor influences.
- Decomposed ultrasonic tool motion into X/Y/Z directional models and reconstructed 3D trajectories via MATLAB.
- Proposed a variable-thickness Y-type cell model and applied energy methods to correlate cell wall mechanical parameters with equivalent properties of honeycomb structures.
- Recorded and processed experimental data using Origin and MATLAB.
- Research Outputs: This project culminated in a master’s thesis, one first-author SCI paper (Q2), and one first-author Chinese core journal paper.

Ultrasonic Cutter System Development for Composite Materials *Oct 2021 - Jun 2023*

- Designed and modeled scalloped disc cutters for ultrasonic machining systems using CREO.
- Familiarized with the design principles of the development of longitudinal-torsional ultrasonic horns through collaboration with lab members.
- Gained exposure to performance studies of ultrasonic transducers under thermo-mechanical coupling effects via peer consultation.

INTERNSHIP & WORK EXPERIENCE

Ningbo Sunny Optical Technology Co., Ltd., Ningbo, China *Aug 2024 - Present*

Opto-Mechanical Structure Design Engineer (Camera Module)

- Proficient in CAD engineering drawing, CREO for 3D modeling, and ABAQUS for finite element simulation.
- Engaged in structural design of mobile phone camera modules, including chip-lens matching, with in-depth understanding of optical lens parameters.
- Led end-to-end design, trial production, and mass production follow-up for the periscope telephoto module used in the OPPO Find X8 flagship model.
- Managed full-process design, trial production, and mass production follow-up for large-image-surface OIS (Optical Image Stabilization) modules.

Hangzhou Zihe Intelligence Co., Ltd., Hangzhou, China *Dec 2023 - Feb 2024*

Assistant Structural Engineer Intern

- Conducted stress analysis and performed generative design for seat and motorcycle sub-components using SolidEdge.
- Assisted structural engineers in drafting motorcycle parts with SolidEdge.

AWARDS & HONORS

Graduate Student Third-Class Scholarship, Hangzhou Dianzi University *Nov 2022*

Graduate Student Second-Class Scholarship, Hangzhou Dianzi University *Nov 2021*

Undergraduate Second-Class Scholarship, Jingdezhen Ceramic Institute *May 2019*